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Front light source module and display apparatus

Abstract

A front light source module is provided to include a side light source; a light guide layer with a light incident side opposite to the side light source in a first direction; and a first light adjusting layer, the first light adjusting layer and the light guide layer are stacked in a third direction, a portion of the first light adjusting layer away from the light guide layer is provided with micro-groove structures, each including: a first inclined surface and a second inclined surface opposite to each other in the first direction, the first inclined surface is configured to face the light incident sid, closer to the light incident side than the second inclined surface, angle α between the first inclined surface and a plane where a surface of the first light adjusting layer away from the light guide layer is located is in a range from 26° to 42°.

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Background/Summary

(1) This is a National Phase Application filed under 35 U.S.C. 371 as a national stage of PCT/CN2023/110351, filed Jul. 31, 2023, an application claiming the benefit of Chinese Application No. 202211183113.0, filed Sep. 27, 2022, the content of each of which is hereby incorporated by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure relates to the field of display technology, and in particular to a front light source module and a display apparatus.
BACKGROUND
(3) At present, under the market trend that the outdoor display and the motion display are more and more favored, panel manufacturers invest most energy in designing a low power consumption display product which may utilize outdoor environment light to display, and a reflective display panel is emerged.
(4) The reflective display panel realizes the display by using a reflective layer (a metal layer with a reflecting function) in the reflective display panel to reflect the ambient light, which does not include a backlight source, and has the advantages of low power consumption, lightweight and thinness and the like. However, when the ambient light is relatively weak, the reflective display product needs an additional light source to assist in the display. Therefore, a development of a front light source module applicable to the reflective display panel has become a new research direction in the display field.
SUMMARY
(5) In a first aspect, embodiments of the present disclosure provide a front light source module, including: a side light source; a light guide layer including a light incident side opposite to the side light source in a first direction; and a first light adjusting layer, wherein the first light adjusting layer and the light guide layer are stacked in a third direction, a portion of the first light adjusting layer away from the light guide layer is provided with a plurality of micro-groove structures, each micro-groove structure includes: a first inclined surface and a second inclined surface opposite to each other in the first direction, the first inclined surface is configured to face the light incident side and is closer to the light incident side than the second inclined surface, an angle α between the first inclined surface and a plane where a surface of the first light adjusting layer away from the light guide layer is located is in a range from 26° to 42°, and a depth H of each micro-groove structure is in a range from 4 μm to 15 μm; wherein a refractive index of the first light adjusting layer is greater than or equal to that of the light guide layer.
(6) In some embodiments, a length of an opening of each micro-groove structure in the surface of the first light adjusting layer away from the light guide layer in the first direction is L.sub.1; and a ratio of L1 to H is L1/H, which satisfies: L.sub.1/H≤4.
(7) In some embodiments, L.sub.1 satisfies L.sub.1≤80 μm.
(8) In some embodiments, the first inclined surface is rectangular, a pair of opposite sides of the rectangular first inclined surface extend along a second direction, and the second direction is perpendicular to both the first direction and the third direction; and an extending direction of the other pair of opposite sides of the rectangular first inclined surface is perpendicular to the third direction, and intersects with both the first direction and the second direction.
(9) In some embodiments, a length L.sub.2 of a side of the rectangular first inclined surface extending in the second direction satisfies: L.sub.2≤80 μm.
(10) In some embodiments, a shape of a cross section of each micro-groove structure taken along a

plane parallel to the first direction and parallel to the third direction includes: a triangle or a quadrangle.

(11) In some embodiments, the second inclined surface is a flat surface or a curved surface.

(12) In some embodiments, a distribution density of the plurality of micro-groove structures increases gradually in a direction away from the light incident side along the first direction starting from the light incident side.

(13) In some embodiments, the surface of the first light adjusting layer away from the light guide layer is divided into a plurality of groove structure arrangement regions arranged along the first direction; a distance between every two adjacent groove structure arrangement regions gradually decreases in the direction away from the light incident side along the first direction starting from the light incident side; each groove structure arrangement region is divided into a plurality of rectangular periodic regions in the second direction; a length of each rectangular periodic region in the first direction is R, and a length of each rectangular periodic region in the second direction is Q; and M micro-groove structures are uniformly arranged in each rectangular periodic region.

(14) In some embodiments, the M micro-groove structures in each rectangular periodic region are arranged to satisfy: a distance between centers of any two micro-groove structures in the first direction is greater than or equal to R/M, and a distance between the centers of any two micro-groove structures in the second direction is greater than or equal to Q/M.

(15) In some embodiments, the side light source includes: a light source; and a converging structure between the light source and the light incident side and configured to converge the light emitted from the light source in the third direction, wherein an angle $\theta 1$ between the light emitted from the converging structure and a first reference plane satisfies: $\theta 1 \leq 52.4°$, and the first reference plane is a plane perpendicular to the third direction.

(16) In some embodiments, the converging structure includes: a wedge-shaped light guiding structure; the wedge-shaped light guiding structure includes: a first light incident surface, a first light outgoing surface, a first light adjusting surface and a second light adjusting surface; the first light incident surface and the first light outgoing surface are opposite to each other in the first direction, both the first light incident surface and the first light outgoing surface are perpendicular to the first direction, a length of the first light incident surface in the third direction is T1, a length of the first light going surface in the third direction is T2, T2>T1, and an orthographic projection of the first light going surface on a plane where the first light incident surface is located covers the first light incident surface; the first light adjusting surface and the second light adjusting surface are opposite to each other in the third direction, and a distance between the first light adjusting surface and the second light adjusting surface in the third direction is gradually increased in a direction from the first light incident surface to the first light outgoing surface along the first direction; and the light source is opposite to the first light incident surface, and the light incident side is opposite to the first light outgoing surface.

(17) In some embodiments, the light source includes: a driving board and a light emitting element fixed on the driving board, a length T of the light emitting element in the third direction is smaller than the length T1 of the first light incident surface in the third direction; and an orthographic projection of the light emitting element on the plane where the first light incident surface is located in a region defined by the first light incident surface.

(18) In some embodiments, the length T of the light emitting element in the third direction satisfies: $T \leq 0.3$ mm.

(19) In some embodiments, the wedge-shaped light guiding structure and the light guide layer are made of a same material and have a one-piece structure; and the light incident side and the first light outgoing surface are a same surface.

(20) In some embodiments, the converging structure includes a condenser lens.

(21) In some embodiments, a shape of a cross section, taken along a plane perpendicular to the second direction, of a surface of the condenser lens away from the light source is a circular arc; or a

shape of a cross section, taken along a plane perpendicular to the second direction, of a surface of the condenser lens away from the light source is a curve formed by connecting circular arcs and line segments sequentially and alternately.

(22) In some embodiments, the light source includes: a driving board and a light emitting element fixed onto the driving board, and the condenser lens is on the driving board and covers the light emitting element.

(23) In some embodiments, the front light source module further includes a plurality of light converging micro-structures on a surface of the light guide layer away from the first light adjusting layer, and configured to converge light passing through the plurality of light converging micro-structures from the light guide layer.

(24) In some embodiments, each light converging micro-structure is a light converging groove on the surface of the light guide layer away from the first light adjusting layer, and the light converging groove extends along the second direction; and a cross section, taken along a plane perpendicular to the second direction Y, of a surface of the light converging groove is V-shaped or arc-shaped.

(25) In some embodiments, a length L.sub.3 of the light converging groove in the first direction satisfies: L.sub.3≤80 μm.

(26) In some embodiments, the front light source module further includes at least one second light adjusting layer on a side of the light guide layer away from the first light adjusting layer, wherein the at least one second light adjusting layer and the light guide layer are stacked in the first direction, and the at least one second light adjusting layer includes light adjusting micro-structures thereon and configured to adjust an exit angle of light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the light adjusting micro-structures.

(27) In some embodiments, the light emitted from the light guide layer away from the first light adjusting layer propagates in a direction away from a plane where the light incident side is located; the light adjusting micro-structures arranged on the at least one second light adjusting layer include: a first light adjusting micro-structure configured so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the first light adjusting micro-structure still propagates in the direction away from the plane where the light incident side is located, but an angle between the light and the third direction increases; and/or the light adjusting micro-structures arranged on the at least one second light adjusting layer include: a second light adjusting micro-structure configured so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the second light adjusting micro-structure still propagates in the direction away from the plane where the light incident side is located, but the angle between the light and the third direction decreases; and/or the light adjusting micro-structures arranged on the at least one second light adjusting layer include: a third light adjusting micro-structure configured so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the third light adjusting micro-structure propagates in a direction close to the plane where the light incident side is located.

(28) In some embodiments, a refractive index of a second light adjusting layer closest to the light guide layer is greater than or equal to the refractive index of the light guide layer.

(29) In some embodiments, the at least one second light adjusting layer includes two or more second light adjusting layers; and a refractive index of one of any two adjacent second light adjusting layers closer to the light guide layer is less than or equal to a refractive index of the other second light adjusting layer.

(30) In some embodiments, the first light adjusting layer is attached to the light guide layer by a first attaching adhesive layer, and a refractive index of the first attaching adhesive layer is greater than or equal to the refractive index of the light guide layer and is less than or equal to the

refractive index of the first light adjusting layer.
(31) In some embodiments, the first light adjusting layer is made of a nano-imprint material.
(32) In a second aspect, an embodiment of the present disclosure further provides a display apparatus, including: a reflective display panel and the front light source module in the first aspect; the front light source module is located on a light outgoing surface of the reflective display panel.
(33) In some embodiments, the reflective display panel includes a plurality of sub-pixel regions arranged in an array along a first direction and a second direction, and each sub-pixel region has a length L.sub.0 in the first direction; and a length of each micro-groove structure in the first direction is less than or equal to ⅔×L.sub.0, and a length of each micro-groove structure in the second direction is less than or equal to ⅔×L.sub.0.
(34) In some embodiments, the display apparatus further includes a plurality of light converging micro-structures on a surface of the light guide layer away from the first light adjusting layer, and configured to converge light passing through the plurality of light converging micro-structures from the light guide layer; wherein a length of each light converging micro-structure in the first direction is less than or equal to ⅔×L.sub.0.
(35) In some embodiments, the display apparatus further includes at least one second light adjusting layer on a side of the light guide layer away from the first light adjusting layer, wherein the at least one second light adjusting layer and the light guide layer are stacked in the first direction, and the at least one second light adjusting layer includes light adjusting micro-structures thereon and configured to adjust an exit angle of light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the light adjusting micro-structures; wherein a length of each light adjusting micro-structure in the first direction is less than or equal to ⅔×L.sub.0.
(36) In some embodiments, the front light source module and the reflective display panel are attached to each other by a second attaching adhesive layer; and a refractive index of a portion of the front light source module in contact with the second attaching adhesive layer is greater than a refractive index of the second attaching adhesive layer.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating a structure of a front light source module and a reflective display panel stacked together according to an embodiment of the present disclosure;
(2) FIG. **2** is a schematic diagram of a structure of a front light source module according to an embodiment of the present disclosure;
(3) FIG. **3** is a schematic cross-sectional view of a front light source module according to an embodiment of the present disclosure;
(4) FIG. **4** is a schematic diagram illustrating how to test reflectivity of a reflective display panel under incident light at different angles;
(5) FIG. **5** is a graph illustrating reflectivity of a reflective display panel under incident light at different angles;
(6) FIG. **6** is a schematic diagram illustrating optical paths at some positions in a front light source module according to an embodiment of the present disclosure;
(7) FIG. **7** is a schematic diagram illustrating light leakage at an opening of a micro-groove structure according to an embodiment of the present disclosure;
(8) FIG. **8** is a schematic diagram illustrating a light leakage probability P at an opening of a micro-groove structure varying with L.sub.1/H when θ.sub.1 is respectively 56.7° and 43° with α=37° according to an embodiment of the present disclosure;
(9) FIG. **9** is a schematic diagram illustrating a light leakage probability P at an opening of a micro-

groove structure varying with α when L.sub.1/H is 11.2/6 according to an embodiment of the present disclosure;
(10) FIG. **10** is a schematic diagram illustrating distribution of the brightness of light emitted from a side of a first light adjusting layer away from a light guide layer at different angles of α which are simulated according to an embodiment of the present disclosure;
(11) FIG. **11** is a schematic diagram illustrating distribution of the brightness of light emitted from a side of a first light adjusting layer close to a light guide layer at different angles of α which are simulated according to an embodiment of the present disclosure;
(12) FIG. **12** is a schematic diagram of various configurations for a micro-groove structure according to an embodiment of the present disclosure;
(13) FIG. **13** is a schematic diagram of a layout of micro-groove structures according to an embodiment of the present disclosure;
(14) FIG. **14** is a schematic diagram illustrating various layouts for four micro-groove structures within a rectangular periodic region according to an embodiment of the present disclosure;
(15) FIG. **15**A is a schematic cross-sectional view of a side light source according to an embodiment of the present disclosure;
(16) FIG. **15**B is another schematic cross-sectional view of a side light source according to an embodiment of the present disclosure;
(17) FIG. **16**A is another schematic cross-sectional view of a side light source according to an embodiment of the present disclosure;
(18) FIG. **16**B is a schematic diagram of a structure of the side light source shown in FIG. **16**A;
(19) FIG. **16**C is a schematic diagram of a light emission effect of the side light source shown in FIG. **16**A;
(20) FIG. **17**A is a yet another schematic cross-sectional view of a side light source according to an embodiment of the present disclosure;
(21) FIG. **17**B is a schematic diagram of a structure of the side light source shown in FIG. **17**A;
(22) FIG. **17**C is a schematic diagram of a light emission effect of the side light source shown in FIG. **17**A;
(23) FIG. **18**A is a schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure;
(24) FIG. **18**B is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure;
(25) FIG. **19**A is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure;
(26) FIG. **19**B is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure;
(27) FIG. **20** is a schematic diagram of another structure of a front light source module according to an embodiment of the present disclosure;
(28) FIG. **21** is a schematic cross-sectional view of another front light source module according to an embodiment of the present disclosure;
(29) FIG. **22** is a schematic diagram illustrating modulation of light by three different light adjusting micro-structures according to an embodiment of the present disclosure;
(30) FIG. **23** is a schematic cross-sectional view of a portion of a display apparatus according to an embodiment of the present disclosure; and
(31) FIG. **24** is another schematic cross-sectional view of a portion of a display apparatus according to an embodiment of the present disclosure.

DETAIL DESCRIPTION OF EMBODIMENTS

(32) In order to enable one of ordinary skill in the art to better understand the technical solutions of the present disclosure, a front light source module and a display apparatus provided by the present disclosure will be described in further detail with reference to the accompanying drawings.

(33) Unless defined otherwise, technical or scientific terms used herein shall have the ordinary meaning as understood by one of ordinary skill in the art to which the present disclosure belongs. The terms “first”, “second”, and the like used in the present disclosure are not intended to indicate any order, quantity, or importance, but rather are used for distinguishing one element from another. Further, the term “a”, “an”, “the”, or the like used herein does not denote a limitation of quantity, but rather denotes the presence of at least one element. The term “comprising”, “including”, or the like means that the element or item preceding the term contains the element or item listed after the term and its equivalent, but does not exclude other elements or items. The terms “upper”, “lower”, “left”, “right”, and the like are used only for indicating relative positional relationships, and when the absolute position of an object being described is changed, the relative positional relationships may also be changed accordingly.
(34) FIG. **1** is a schematic diagram illustrating a structure of a front light source module and a reflective display panel stacked together according to an embodiment of the present disclosure; FIG. **2** is a schematic diagram of a structure of a front light source module according to an embodiment of the present disclosure; FIG. **3** is a schematic cross-sectional view of a front light source module according to an embodiment of the present disclosure. As shown in FIG. **1** to FIG. **3**, a front light source module may be disposed on a light outgoing side of a reflective display panel **100** to provide light for display (display light) to the reflective display panel **100** in a weak ambient light scene. The front light source module includes: a side light source **400**, a light guide layer **340** and a first light adjusting layer **320**.
(35) The light guide layer **340** includes a light incident side **341**, and the light incident side **341** and the side light source **400** are disposed opposite to each other in a first direction X (e.g., a horizontal direction in FIG. **3**). The light from the side light source **400** may be incident into the light guide layer **340** through the light incident side **341**. The first light adjusting layer **320** and the light guide layer **340** are stacked in a third direction Z (e.g., a vertical direction in FIG. **3**), a side of the first light adjusting layer **320** away from the light guide layer **340** is provided with a plurality of micro-groove structures **310**, and each micro-groove structure **310** includes: a first inclined surface **311** and a second inclined surface **312** opposite to each other in the first direction X. The first inclined surface **311** is configured to face the light incident side **341** and is closer to the light incident side **341** than the second inclined surface **312**, an angle α between the first inclined surface **311** and a plane where a surface of the first light adjusting layer **320** away from the light guide layer **340** is located is in a range from 26° to 42°, and a depth H of each micro-groove structure **310** is in a range from 4 μm to 15 μm. A refractive index of the first light adjusting layer **320** is greater than or equal to that of the light guide layer **340**.
(36) In some embodiments, the first light adjusting layer **320** is attached by a first attaching adhesive layer **330**, a refractive index of the first attaching adhesive layer **330** is greater than or equal to that of the light guide layer **340** and is less than or equal to that of the first light adjusting layer **320**.
(37) In the embodiment of the present disclosure, the light provided by the side light source **400** may be incident into the light guide layer **340** through the light incident side **341**, and part of the light may be incident into the first light adjusting layer **320** during the light propagation process in the light guide layer **340**. The micro-groove structures **310** are disposed in the first light adjusting layer **320**, each micro-groove structure **310** includes the first inclined surface **311** facing the light incident side **341**, and the first inclined surface **311** may reflect the part of the light incident into the first light adjusting layer **320** toward the reflective display panel **100** (i.e., a side of the first light adjusting layer **320** close to the light guide layer **340**), so as to provide light for display to the reflective display panel **100**.
(38) The refractive index of the first light adjusting layer **320** is greater than or equal to that of the light guide layer **340**, so that total reflection of light occurring when the light is incident from the light guide layer **340** to the first light adjusting layer **320** can be effectively avoided, the quantity of

light which may be transmitted to the first light adjusting layer **320** is ensured, and the quantity of light which are finally provided to the reflective display panel **100** by the front light source module is favorably improved.
(39) Similarly, with the first attaching adhesive layer **330**, the refractive index of the first attaching adhesive layer **330** is greater than or equal to the refractive index of the light guide layer **340** and less than or equal to the refractive index of the first light adjusting layer **320**, the total reflection of light occurring when the light is incident from the light guide layer **340** to the first light adjusting layer **320** can be also avoided.
(40) In addition, in the embodiment of the present disclosure, the light source **400** is unnecessarily turned on in an environment with a strong ambient light, and the display may be realized by the ambient light. At this time, the ambient light needs to pass through the front light source module to the reflective display panel **100**, and forms the light for imaging (imaging light) by light reflection of the reflective display panel **100**, and the imaging light passes through the front light source module and then exits. In the procedure that the external ambient light or the imaging light passes through the front light source module, the micro-groove structures **310** in the first light adjusting layer **320** will certainly affect the external ambient light or the imaging light, which will affect the final display quality. To solve the above problem, a size of each micro-groove structure **310** in the first light adjusting layer **320** is designed to be smaller in the embodiment of the present disclosure. For example, a depth H of each micro-groove structure **310** is in a range from 4 μm to 15 μm. However, since the size of the micro-groove structure **310** is too small and the requirements for the flatness and the inclined angle accuracy of the first inclined surface are high, it is difficult to form the micro-groove structures **310** by a conventional patterning process (it is difficult to form the flat first inclined surface and control a first inclined angle by the patterning process) or an injection molding process (it is difficult to form the micro-groove structures with a small size due to the process accuracy of the injection molding process). Therefore, in the embodiment of the present disclosure, a material of the first light adjusting layer **320** may be a nano-imprint material. At this time, the micro-groove structures **310** with a small size may be formed by a nano-imprint process, and the first inclined surface of each formed micro-groove structure **310** is flat. In some embodiments, the nano-imprint material includes nano-imprint glue, such as acrylic resin. A refractive index of the nano-imprint material may be adjusted by adding inorganic particles (e.g., $TiO2$, $ZrO2$, etc.) to the nano-imprint glue.
(41) In addition, in the embodiment of the present disclosure, the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is in a range from 26° to 42°, so that an incident angle of the display light reflected by the first inclined surface **311** and emitted to the reflective display panel **100** is within a high-reflectivity incident angle range of the reflective display panel **100** as much as possible when the display light reaches the reflective display panel **100**, and an exit angle of the light refracted on the first inclined surface **311** and finally emitted from an opening of each micro-groove structure **310** is outside a viewing angle range of a display apparatus as much as possible.
(42) FIG. **4** is a schematic diagram illustrating a method for testing reflectivity of a reflective display panel under incident light at different angles; FIG. **5** is a graph illustrating reflectivity of a reflective display panel under incident light at different angles. As shown in FIGS. **4** and **5**, a reflective layer in the reflective display panel **100** mainly performs a specular reflection, and a scattering layer or a panel bump structure in the reflective display panel **100** modulates the light reflected by the reflective layer to a main viewing angle. In the test scene shown in FIG. **4**, a light detection unit is aligned with (directly faces) the reflective display panel **100**, and then a position of a test light source is continuously adjusted to change an angle θ of the light emitted to the reflective display panel **100**. The angle θ may vary in a range from −90° to 90°.
(43) It is found through testing that when the light is incident on the reflective display panel **100** at different angles θ, a reflectivity of the reflective display panel **100** is greatly different. The

reflective display panel **100** may exhibit a certain reflectivity when the angle θ of the incident light is between −45° and +45°, and the reflective display panel **100** exhibits a higher reflectivity (greater than 40%) when the angle θ of the incident light is between −10° and −30°.
(44) In the present disclosure, an angle α between the first inclined surface **311** and a plane where a surface of the first light adjusting layer **320** away from the light guide layer **340** is located directly affects an incident angle of the display light reflected by the first inclined surface **311** when the display light reaches the reflective display panel **100**.
(45) FIG. **6** is a schematic diagram of optical paths at some positions in a front light source module according to an embodiment of the present disclosure. As shown in FIG. **6**, the refractive index of the light guide layer **340** is denoted by n(**340**), the refractive index of the first attaching adhesive layer **330** is denoted by n(**330**), the refractive index of the first light adjusting layer **320** is denoted by n(**310**), refractive indices of a medium in contact with a surface of the first light adjusting layer **320** away from the light guide layer **340** and a medium in contact with the light incident side **341** are denoted by n(air), and n(air)<n(**340**)<n(**330**)<n(**310**). An angle between the second inclined surface **312** and a surface of the first light adjusting layer **320** away from the light guide layer **340** is α.sub.2, and a length of the opening of each micro-groove structure **310** in the first direction X is L.sub.1.
(46) A critical angle of total reflection of the surface of the first light adjusting layer **320** away from the light guide layer **340** is denoted by θ.sub.c1:
(47) $_{c1} = \arcsin(\frac{n(\text{air})}{n(310)})$
(48) A critical angle of total reflection at an interface between the first light adjusting layer **320** and the first attaching adhesive layer **330** is denoted by θ.sub.c2:
(49) $_{c2} = \arcsin(\frac{n(330)}{n(310)})$
(50) The following four optical paths may exist for the light input from the light incident side **341** to the first light adjusting layer **320**:
(51) Optical path (a): the light reaches the surface of the first light adjusting layer **320** away from the light guide layer **340** at an angle θ.sub.5, and then is reflected by the surface, and the reflected light does not pass through the micro-groove structures **310**. At this time, θ.sub.5<90−α, the reflected light travels downward at the angle θ.sub.5. At this time, the following relationships exist:
(52) $n(\text{air}) \times \sin\ _1 = n(340) \times \sin\ _2$ Formula(1) $_3 = 90 -\ _2$
$n(340) \times \sin\ _3 = n(330) \times \sin\ _4 = n(310) \times \sin\ _5$ $_5 = \arcsin(\frac{n(340)}{n(310)} \times \sqrt{1 - (\frac{n(\text{air})}{n(340)})^2 \times (\sin\ _1)^2})$
(53) Optical path (b): the light reaches the surface of the first light adjusting layer **320** away from the light guide layer **340** at the angle θ.sub.5, and then is reflected by the surface, and the reflected light reaches a first inclined surface **311** of a micro-groove structure **310** and is reflected by first inclined surface **311**. In order to realize the light convergence, an angle k of the reflected light is less than θ.sub.5. At this time, the angle k of the reflected light meets the following relationship:
(54) $k = 180 -\ _5 - 2 \times$ Formula(2)
(55) Optical path (c): the light is directly irradiated and is refracted on the first inclined surface **311** of the micro-groove structure **310**, and the refracted light is directly emitted from the opening of the micro-groove structure **310**, that is, light leakage is generated at the micro-groove structure **310**. At this time, an exit angle δ of the refracted light meets the following relationship:
(56) $= + \arcsin(\frac{n(310)}{n(\text{air})} \times \sin(\ _5 -))$ Formula(3)
(57) Optical path (d): the light is irradiated and is refracted on the first inclined surface **311** of the micro-groove structure **310**, but the refracted light travels to the second inclined surface **312** and is refracted on the second inclined surface **312** and enters the first light adjusting layer **320** again. At this time, a refraction angle ω of the refracted light meets the following relationship:
(58) $_2 = \arctan(\frac{H}{L_1 - H \times \cot})$ Formula(4) $= \ _2 + \arcsin(\frac{n(\text{air})}{n(310)} \times \sin(180 - - \ _2))$

$$= \arctan\left(\frac{H}{L_1 - H \times \cot}\right) + \arcsin\left(\frac{n(\text{air})}{n(310)} \times \sin(180 - \quad - (\arctan(\frac{H}{L_1 - H \times \cot})))\right)$$

(59) Since the refraction angle ω is large, most of the light may be constrained to be totally reflected by the first light adjusting layer **320**. In order to destroy the total reflection and cause the light to continuously propagate downward, it is necessary to increase the critical angle θ.sub.c2 of total reflection at the interface between the first light adjusting layer **320** and the first attaching adhesive layer **330**. Preferably, the refractive index n(**330**) of the first attaching adhesive layer **330** is equal to the refractive index n(**310**) of the first light adjusting layer **320**, and at this time, the light with any refraction angle ω may pass through the interface between the first light adjusting layer **320** and the first adhesive layer.

(60) To facilitate a better understanding of the technical solution of the present disclosure for one of ordinary skill in the art, a detailed description will be given below with reference to a specific example.

(61) Where n(air) is 1, n(**340**) is 1.49, n(**330**) is 1.55, n(**310**) is 1.58, α is 37°, L.sub.1 is 11.2 μm, and H is 6 μm. The following table 1 may be obtained by calculation using the above formulas (1) to (4).

(62) TABLE-US-00001 TABLE 1 Angle calculation result table for different optical paths Intensity of incident light at Angle calculation different angles Optical Optical Optical Optical at light incident path (a) path (b) path (c) path (d) side 341 θ.sub.1 θ.sub.5 θ.sub.c1 θ.sub.5 k δ ω θ.sub.c2 0 90° 44.4°.sup. 39.3° 44.4°.sup. — 48.8°.sup. 98° 78.8° 0.3 × I.sub.0 70° 47° 39.3° 47° — 53° 96.8° 78.8° 0.5 × I.sub.0 60° 50° 39.3° 50° — 58° 95° 78.8° 0.6 × I.sub.0 52.4°.sup. 53° 39.3° 53° 53° 62.8°.sup. 93° 78.8° 0.7 × I.sub.0 43° 57° 39.3° — 49° 70° 90° 78.8° 0.95 × I.sub.0 19° 67° 39.3° — 39° 89° 79.6° 78.8° 0.98 × I.sub.0 12° 69° 39.3° — 37° — 76.9° 78.8° I.sub.0 0° 70.6°.sup. 39.3° — 35.4°.sup. — 74.4° 78.8°

(63) In the above table 1, the intensity of the light incident on the light incident side **341** at the incident angle θ.sub.1=0° is I.sub.0, the intensity of the light incident on the light incident side **341** at the incident angle θ.sub.1=12° is 0.98×I.sub.0, the intensity of the light incident on the light incident side **341** at the incident angle θ.sub.1=19° is 0.95×I.sub.0, the intensity of the light incident on the light incident side **341** at the incident angle θ.sub.1=43° is 0.7×I.sub.0, and so on.

(64) As can be seen from the results in Table 1, the angle ω of the refracted light for the optical path (d) is large (ω≥74.4°). Even if such the refracted light with the large angle may be transmitted to the reflective display panel, it can be seen from a reflectivity curve in FIG. **5** that the reflectivity of the reflective display panel for the light with the large angle is extremely low. Therefore, the utilization value for the light in the optical path (d) is low.

(65) The angles of the reflected light formed in the optical paths (a) and (b) are less than or equal to 53°, and thus the light is the light with the small angle. It can be seen from the reflectivity curve in FIG. **5** that the reflectivity of the reflective display panel for the light with the small angle is relatively high. Therefore, the utilization value for the light in the optical paths (a) and (b) is relatively high. Further, the reflected light formed in the optical path (a) mainly comes from the light with the incident angle θ.sub.1 in a range from 52.4° to 90° at the light incident side **341**, the reflected light formed in the optical path (b) mainly comes from the light with the incident angle θ.sub.1 in a range from 0° to 52.4° at the light incident side **341**, and the intensity of the light with the incident angle θ.sub.1 in a range from 0° to 52.4° is significantly greater than the intensity of the light with the incident angle θ.sub.1 in a range from 52.4° to 90°, that is, the intensity of the reflected light for display formed in the optical path (b) is greater than that in the optical path (a). That is, the utilization value for the light in the optical path (b) is higher than that in the optical path (a).

(66) Based on the above calculation results, it can be concluded that when the light output by the side light source **400** is converged so that the incident light angle θ.sub.1 of all the light incident to the light incident side **341** is in a range from 0° to 52.4°, the quantity of light forming the optical path (b) can be effectively increased, so as to increase the quantity of light provided by the front

light source module to the reflective display panel **100**, which is beneficial to increasing the display brightness.
(67) Due to the existence of the optical path (c), which indicates that there is light leakage at the micro-groove structure **310**, the light leakage at the micro-groove structure **310** directly affects the contrast of the display apparatus. For this reason, an exit angle δ of the light leaked from the micro-groove structure **310** should be larger than a maximum viewing angle δ max of the display apparatus as much as possible, so as to prevent the user from receiving the light leakage at the micro-groove structure **310** under the viewing angle. Generally, the maximum viewing angle δ max of the display apparatus is 60°.
(68) From the formulas (1) and (3), it may be obtained that θ.sub.1 satisfies the following relationship:
(69) Formula5 $_1 = \arcsin\sqrt{\left(\frac{n(340)}{n(\mathrm{air})}\right)^2 \times \left[1 - \left(\frac{n(310)}{n(340)} \times \sin\left(\quad + \arcsin\left(\frac{n(\mathrm{air})}{n(310)} \times \sin(\quad - \quad)\right)\right)\right)^2\right]}$
(70) When δ>δ max=60°, the following relationship may be obtained:
(71) $_1 < \arcsin\sqrt{\left(\frac{n(340)}{n(\mathrm{air})}\right)^2 \times \left[1 - \left(\frac{n(310)}{n(340)} \times \sin\left(\quad + \arcsin\left(\frac{n(\mathrm{air})}{n(310)} \times \sin(60^\circ - \quad)\right)\right)\right)^2\right]}$
(72) By substituting n(air)=1, n(**340**)=1.49, n(**310**)=1.58, α=37° into Formula (5), θ.sub.1 <56.7° may be obtained through calculation.
(73) It should be noted that when δ max is larger, the exit angle δ of the leaked light will be further limited. For example, when the maximum viewing angle δ max is 70°, θ.sub.1<43° may be obtained by calculation. Therefore, in order to further reduce the effect of the light leakage at the opening of the micro-groove structure **310** with the light leakage, the light with the incident light angle θ.sub.1 at the light incident side **341** may be further converged.
(74) FIG. **7** is a schematic diagram illustrating light leakage at an opening of a micro-groove structure according to an embodiment of the present disclosure. As shown in FIG. **7**, the refracted light generated after being irradiated to a region indicated by w′ forms light leakage, and the refracted light generated after being irradiated to a region indicated by w-w′ enters the first light adjusting layer **320** again through the second inclined surface **312** (no light leakage is formed).
(75) A light leakage probability at the opening of the micro-groove structure **310** is defined as P, and P satisfies:
(76) $P = \frac{w'}{w} = \frac{L1}{H \times (\cot \quad + \tan \quad)} = \frac{L1}{H} \times \frac{1}{(\cot \quad + \tan \quad)}$ Formula(6)
(77) As can be seen from Formula (6), the light leakage probability P is related to L1, H, α, and δ.
(78) FIG. **8** is a schematic diagram illustrating a light leakage probability P at an opening of a micro-groove structure **310** varying with L.sub.1/H when θ.sub.1 is respectively 56.7° and 43°, with α=37° according to an embodiment of the present disclosure. As shown in FIG. **8**, α is 37°. When θ.sub.1 is 56.7°, the light leakage probability is linearly increased along with the increase of L.sub.1/H; when θ.sub.1 is 43°, the light leakage probability is linearly increased along with the increase of L.sub.1/H.
(79) FIG. **9** is a schematic diagram illustrating a light leakage probability P at an opening of a micro-groove structure **310** varying with α when L.sub.1/H is 11.2/6 according to an embodiment of the present disclosure. As shown in FIG. **9**, L.sub.1/H is 11.2/6. With α in a range from 20° to 45°, the light leakage probability increases along with the increase of α.
(80) FIG. **10** is a schematic diagram illustrating brightness distribution of light from a side of a first light adjusting layer away from a light guide layer simulated at different angles α according to an embodiment of the present disclosure. FIG. **11** is a schematic diagram illustrating brightness distribution of light from a side of a first light adjusting layer **320** close to a light guide layer **340** simulated at different angles of α according to an embodiment of the present disclosure. As shown in FIGS. **10** and **11**, it is defined that the light forms a negative (−) angle with the third direction Z when propagating in a direction away from the light incident side **341**, and the light forms a positive (+) angle with the third direction Z when propagating in a direction close to the light

incident side **341** (in the same manner as the angle θ in FIGS. **4** and **5**).
(81) As can be seen from the result of FIG. **10**, when α is 32°, 37° or 42°, an angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** away from the light guide layer **340** is greater than 75°, and the exit angle of most of the light is greater than 60°. That is, the exit angle of most of the light leaking from the opening of the micro-groove structure **310** is outside the viewing angle range of the display apparatus (the maximum viewing angle δ max is typically 60°). Therefore, when α is set in a range from 32° to 42°, the exit angle of the light refracted on the first inclined surface **311** and finally emitted from the opening of each micro-groove structure **310** is outside the viewing angle range of the display apparatus as much as possible, which is beneficial to improving the contrast of the display apparatus.
(82) As can be seen from the results of FIG. **11**, when α is 32°, an angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** close to the light guide layer **340** is −34°. When α is 37°, an angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** close to the light guide layer **340** is −21°. When α is 42°, an angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** close to the light guide layer **340** is −16°. Under three different a values, the angles corresponding to a peak brightness of the light outgoing are better matched with the high-reflectivity incident angle range (from −10° to −30°) of the reflective display panel **100** shown in FIG. **5**.
(83) Therefore, in the embodiment of the present disclosure, the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is in a range from 32° to 42°, so that the incident angle of the display light reflected by the first inclined surface **311** and emitted to the reflective display panel **100** is within the high-reflectivity incident angle range of the reflective display panel **100** as much as possible (to improve the display brightness) when the display light reaches the reflective display panel **100**, and the exit angle of the light refracted on the first inclined surface **311** and finally emitted from an opening of each micro-groove structure **310** is outside the viewing angle range of the display apparatus as much as possible (to improve display contrast).
(84) It should be noted that when the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is smaller than 32° and greater than or equal to 26°, the simulated angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** away from the light guide layer **340** is still greater than 60°, and the angle corresponding to a peak brightness of the light outgoing of a side of the first light adjusting layer **320** close to the light guide layer **340** is about −40°, that is, the exit angle of most of the light emitted from the opening of the micro-groove structure **310** is outside the viewing angle range of the display apparatus, and the display light reflected by the first inclined surface **311** and emitted to the reflective display panel **100** is better matched with the high-reflectivity incident angle range of the reflective display panel **100**.
(85) When the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is greater than 42°, as can be seen from FIG. **9**, the light leakage probability is rapidly increased to a higher level of 50%, which is not favorable for improving the display contrast.
(86) Based on the above comprehensive consideration, in the embodiment of the present disclosure, the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is in a range from 26° to 42°.
(87) In addition, in the embodiment of the present disclosure, as shown in FIG. **8**, the larger L.sub.1/H is, the larger the light leakage probability is. In the embodiment of the present disclosure, in order to effectively control the light leakage probability of the micro-groove structure **310**, L.sub.1/H satisfies: L.sub.1/H≤4. In practical applications, L.sub.1 may be designed to be as small as possible under the conditions of the process operation. Optionally, L.sub.1 satisfies L.sub.1≤80

μm.
(88) FIG. **12** is a schematic diagram of various configurations of a micro-groove structure according to an embodiment of the present disclosure. As shown in FIG. **12**, in some embodiments, each first inclined surface **311** is rectangular, and a pair of opposite sides of the rectangular first inclined surface **311** extend along the second direction Y perpendicular to both the first direction X and the third direction Z. An extending direction of the other pair of opposite sides of the rectangular first inclined surface **311** is perpendicular to the third direction Z, and intersects with both the first direction X and the second direction Y. Optionally, the first direction X, the second direction Y, and the third direction Z are perpendicular to each other.
(89) In some embodiments, a length L.sub.2 of each side of the rectangular first inclined surface **311** extending in the second direction Y satisfies: L.sub.2≤80 μm.
(90) In some embodiments, a shape of a cross section of each micro-groove structure **310** taken along a plane parallel to the first direction X and parallel to the third direction Z includes: a triangle or a quadrangle. For example, the shape of the cross section of the micro-groove structure **310** taken along a plane parallel to the first direction X and parallel to the third direction Z as shown in parts (a) and (d) in FIG. **12** is a triangle, the shape of the cross section of the micro-groove structure **310** taken along a plane parallel to the first direction X and parallel to the third direction Z as shown in parts (b) and (e) in FIG. **12** is a trapezoid, and the shape of the cross section of the micro-groove structure **310** taken along a plane parallel to the first direction X and parallel to the third direction Z as shown in a part (c) in FIG. **12** is a parallelogram.
(91) In some embodiments, each second inclined surface **312** is a flat surface or a curved surface. For example, the second inclined surface **312** is a flat surface as shown in parts (a), (b), and (c) in FIG. **12**, and the second inclined surface **312** is a curved surface as shown in parts (d) and (e) in FIG. **12**.
(92) It should be noted that in the embodiment of the present disclosure, the micro-groove structure **310** may also adopt other shapes as long as the angle α between the first inclined surface **311** and the plane where the surface of the first light adjusting layer **320** away from the light guide layer **340** is located is in a range from 26° to 42°, and a depth H of the micro-groove structure **310** is in a range from 4 μm to 15 μm.
(93) FIG. **13** is a schematic diagram illustrating a layout of micro-groove structures **310** according to an embodiment of the present disclosure. As shown in FIG. **13**, in a direction away from the light incident side **341** along the first direction X starting from the light incident side **341**, a distribution density of the micro-groove structures **310** is gradually increased, which is favorable for improving uniformity of light from the front light source module to the reflective display panel **100**, and thus is favorable for improving brightness uniformity of an image displayed by the display apparatus.
(94) In some embodiments, the surface of the first light adjusting layer **320** away from the light guide layer **340** is divided into a plurality of groove structure arrangement regions **31**P arranged along the first direction X. A distance between every two adjacent groove structure arrangement regions **31**P gradually decreases in the direction away from the light incident side **341** along the first direction X starting from the light incident side **341**. Each groove structure arrangement region **31**P is divided into a plurality of rectangular periodic regions **31**G arranged in the second direction Y. A length of each rectangular periodic region **31**G in the first direction X is R, and a length of each rectangular periodic region **31**G in the second direction Y is Q. M micro-groove structures **310** are uniformly arranged in each rectangular periodic region **31**G.
(95) Further alternatively, the M micro-groove structures **310** in each rectangular periodic region **31**G are arranged to satisfy: a distance between centers of any two micro-groove structures **310** in the first direction X is greater than or equal to R/M, and a distance between the centers of any two micro-groove structures **310** in the second direction Y is greater than or equal to Q/M. With this arrangement, a distance in the first direction X between the centers of any two micro-groove structures **310** closest to each other in the first direction X in each rectangular periodic region **31**G

is equal to R/M, and a distance in the second direction Y between the centers of any two micro-groove structures **310** closest to each other in the second direction Y in each rectangular periodic region **31**G is equal to Q/M, so that the whole groove structure arrangement region **31**P has better brightness uniformity.

(96) In some embodiments, Q/M is less than or equal to 150 μm, i.e., it is ensured that the distance in the second direction Y between the centers of any two micro-groove structures **310** closest to each other in the second direction Y is not greater than 150 μm. With this arrangement, it can be ensured that the picture is fine and smooth, and has no granular sensation (if the distance between the adjacent micro-groove structures **310** in the second direction is too great, human eyes may recognize the granular sensation generated by the micro-groove structures).

(97) FIG. **14** is a schematic diagram illustrating various layouts of four micro-groove structures within a rectangular periodic region according to an embodiment of the present disclosure. As shown in FIG. **14**, by taking M with a value of four as an example, the distance in the first direction X between the centers of any two micro-groove structures **310** closest to each other in the first direction X in each rectangular periodic region is equal to R/4, and the distance in the second direction Y between the centers of any two micro-groove structures **310** closest to each other in the second direction Y in each rectangular periodic region is equal to Q/4. In this case, a connection line sequentially connecting the centers of the four micro-groove structures **310** may be a straight line or a broken line. For example, the connection line sequentially connecting the centers of the four micro-groove structures **310** is a straight line as shown in a part (a) of FIG. **14**, and the connection line sequentially connecting the centers of the four micro-groove structures **310** is a broken line as shown in parts (b), (c), and (d) of FIG. **14**.

(98) FIG. **15**A is a schematic cross-sectional view of a side light source according to an embodiment of the present disclosure; FIG. **15**B is another schematic cross-sectional view of a side light source according to an embodiment of the present disclosure. As shown in FIGS. **15**A and **15**B, in some embodiments, the side light source **400** includes: a light source **401** and a converging structure, the converging structure is located between the light source **401** and the light incident side **341** and configured to converge the light emitted from the light source in the third direction Z, and an angle θ.sub.1 between the light emitted from the converging structure and a first reference plane satisfies: θ.sub.1≤52.4°, and the first reference plane is a plane perpendicular to the third direction Z.

(99) As can be seen from the table 1, the light emitted from the light source is converged by the converging structure, so that the quantity of light forming the optical path (b) can be effectively increased, the quantity of light provided by the front light source module to the reflective display panel **100** is increased, and the display brightness is favorably improved.

(100) Continuing to refer to FIGS. **15**A and **15**B, in some embodiments, the converging structure includes: a wedge-shaped light guiding structure **402**, which includes: a first light incident surface **4021**, a first light outgoing surface **4022**, a first light adjusting surface **4023** and a second light adjusting surface **4024**, the first light incident surface **4021** and the first light outgoing surface **4022** are oppositely arranged in the first direction X, the first light incident surface **4021** and the first light outgoing surface **4022** are perpendicular to the first direction X, a length of the first light incident surface **4021** in the third direction Z is T1, a length of the first light outgoing surface in the third direction Z is T2, T2>T1, and an orthographic projection of the first light outgoing surface **4022** on a plane where the first light incident surface **4021** is located covers the first light incident surface **4021**, the first light adjusting surface **4023** and the second light adjusting surface **4024** are oppositely arranged in the third direction Z, and a distance between the first light adjusting surface **4023** and the second light adjusting surface **4024** in the third direction Z is gradually increased in a direction from the first light incident surface **4021** to the first light outgoing surface **4022** along the first direction X, the light source is opposite to the first light incident surface **4021**, and the light incident side **341** is opposite to the first light outgoing surface **4022**.

(101) In some embodiments, the light source **401** includes: a driving board **4011** and a light emitting element **4012** fixed on the driving board **4011**, wherein a length T of the light emitting element **4012** in the third direction Z is smaller than the length T1 of the first light incident surface in the third direction Z, an orthographic projection of the light emitting element **4012** on the plane where the first light incident surface is located in a region defined by the first light incident surface.
(102) In some embodiments, the light emitting element **4012** may be an LED chip.
(103) In some embodiments, the length T of the light emitting element **4012** in the third direction Z satisfies: $T \leq 0.3$ mm.
(104) As a specific example, T is 0.2 mm, T1 is 0.25 mm, T2 is 0.4 mm, and a distance LT between the first light incident surface and the first light outgoing surface in the first direction X is 1.2 mm.
(105) In some embodiments, referring to FIG. **15**A, the first light adjusting surface and the second light adjusting surface are both flat surfaces. In other embodiments, referring to FIG. **15**B, the first light adjusting surface and the second light adjusting surface are both curved surfaces.
(106) In some embodiments, the wedge-shaped light guiding structure **402** and the light guide layer **340** are made of the same material and have a one-piece structure, and the light incident side **341** and the first light outgoing surface are the same surface.
(107) FIG. **16**A is another schematic cross-sectional view of a side light source **400** according to an embodiment of the present disclosure; FIG. **16**B is a schematic diagram of a structure of the side light source **400** shown in FIG. **16**A; FIG. **16**C is a schematic diagram of a light emission effect of the side light source **400** shown in FIG. **16**A; FIG. **17**A is a yet another schematic cross-sectional view of a side light source **400** according to an embodiment of the present disclosure; FIG. **17**B is a schematic diagram of a structure of the side light source **400** shown in FIG. **17**A; FIG. **17**C is a schematic diagram of a light emission effect of the side light source **400** shown in FIG. **17**A. As shown in FIGS. **16**A to **17**C, in some embodiments, the converging structure includes a condenser lens **403**.
(108) In some embodiments, optionally, the condenser lens **403** is made of a resin material, may be formed on the driving board **4011** by an injection molding process, and covers the light emitting element **4012**.
(109) Referring to FIGS. **16**A and **16**B, as an alternative embodiment, the condenser lens **403** is a cylindrical lens (extending in the second direction Y), and a shape of a cross section of a surface of the condenser lens **403** away from the light source in a direction perpendicular to the second direction Y is a circular arc.
(110) As a specific example, the shape of the cross section of the condenser lens **403** shown in FIG. **16**A in the direction perpendicular to the second direction Y includes a fixing portion **4031**, an intermediate portion **4032**, and a light adjusting portion **4033**, the fixing portion **4031** is located on two opposite sides of the light emitting element in the third direction Z, the intermediate portion **4032** is located between the fixing portion **4031** and the light adjusting portion **4033**, the intermediate portion **4032** is rectangular in shape, an edge of the light adjusting portion **4033** close to the intermediate portion **4032** is a line segment in shape, and an edge of the light adjusting portion **4033** away from the intermediate portion **4032** is a circular arc in shape. A length of the intermediate portion **4032** in the third direction Z is H′, a length of the intermediate portion **4032** in the first direction X is L′, a midpoint of the edge, which is the line segment in shape, of the light adjusting portion **4033** close to the intermediate portion **4032** is denoted as a point O, and a center of a circle of the edge, which is a circular arc in shape, of the light adjusting portion **4033** away from the intermediate portion **4032** is denoted as a point O′, and a radius thereof is denoted as R. As a specific scheme, H′ is 0.4 mm, L′ is 0.1 mm, the point O overlaps with the point O′, and R is 0.2 mm.
(111) As another alternative, referring to FIGS. **17**A and **17**B, a surface of the condenser lens **403** away from the light source is a curved surface formed by arranging a plurality of arc surfaces along the third direction. In this case, the condenser lens **403** is a fresnel lens. A thickness of the

condenser lens **403** shown in FIGS. **17**A and **17**B is smaller than that of the condenser lens **403** shown in FIGS. **16**A and **16**B.

(112) As a specific example, the shape of the cross section of the surface of the condenser lens **403** away from the light source in the direction perpendicular to the second direction Y is a curve formed by connecting circular arcs and line segments sequentially and alternately, and the curve is an axisymmetric pattern (with respect to a symmetry axis parallel to the first direction X).

(113) As a specific example, the shape of the cross section of the condenser lens **403** shown in FIG. **17**A taken along a plane perpendicular to the second direction Y includes the fixing portion **4031**, the intermediate portion **4032**, and the light adjusting portion **4033**, the fixing portion **4031** is located on two opposite sides of the light emitting element in the third direction Z, the intermediate portion **4032** is located between the fixing portion **4031** and the light adjusting portion **4033**, the intermediate portion **4032** is rectangular in shape, an edge of the light adjusting portion **4033** close to the intermediate portion **4032** is a line segment in shape, and an edge of the light adjusting portion **4033** away from the intermediate portion **4032** is a curve formed by connecting circular arcs and line segments sequentially and alternately, a complete circular arc may be formed by translating in the first direction X and sequentially connecting all the circular arcs in the curve. A length of the intermediate portion **4032** in the third direction Z is H′, a length of the intermediate portion **4032** in the first direction X is L′, a midpoint of the edge, which is the line segment in shape, of the light adjusting portion **4033** close to the intermediate portion **4032** is denoted as a point O, and a center of a circle of the edge, which is a circular arc in shape and in the middle, of the light adjusting portion **4033** away from the intermediate portion **4032** is denoted as a point O′, and a radius thereof is denoted as R. Furthermore, the edge of the light adjusting portion **4033** away from the intermediate portion **4032** includes 7 circular arcs and 6 line segments which are alternately arranged in the second direction Y, wherein the 7 circular arcs are sequentially arranged in the second direction Y, a first circular arc and a seventh circular arc are axisymmetric, a second circular arc and a sixth circular arc are axisymmetric, a third circular arc and a fifth circular arc are axisymmetric, and a fourth circular arc is axisymmetric. A length of the i.sup.th circular arc in the third direction Z is denoted as H.sub.i, and i is any of 1 to 7. Specifically, H′ is 0.4 mm, L′ is 0.1 mm, the point O′ is located on the side of the point O away from the light adjusting portion **4033**, a line connecting the point O′ and the point O is parallel to the first direction X, R is 0.2 mm, H.sub.1=H.sub.7=11 μm, H.sub.2=H.sub.6=33 μm, H.sub.3=H.sub.5=66 μm, and H.sub.4=340 μm.

(114) Alternatively, the condenser lens **403** in the embodiment of the present disclosure may also have other shapes, which is not illustrated here.

(115) FIG. **18**A is a schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure; FIG. **18**B is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure; FIG. **19**A is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure; FIG. **19**B is another schematic cross-sectional view of a light guide layer according to an embodiment of the present disclosure. As shown in FIGS. **18**A to **19**B, in some embodiments, a plurality of light converging micro-structures **342** are disposed on a surface of the light guide layer **340** away from the first light adjusting layer **320**, and are configured to converge light passing through the light converging micro-structures **342** from the light guide layer **340**.

(116) In some embodiments, each light converging micro-structure **342** is a light converging groove formed on the surface of the light guide layer **340** away from the first light adjusting layer **320**, and the light converging groove extends along the second direction Y. A cross section of a surface of the light converging groove along a plane perpendicular to the second direction Y is V-shaped or arc-shaped. For example, the cross section of the surface of the light converging groove shown in FIGS. **18**A and **18**B is V-shaped, and the cross section of the surface of the light converging groove shown in FIGS. **19**A and **19**B is arc-shaped.

(117) It should be noted that in the embodiment of the present disclosure, any two adjacent light

converging grooves may be in contact with each other (for example, as shown in FIG. **18**A or FIG. **19**A), or may be arranged at intervals (for example, as shown in FIG. **18**B or FIG. **19**B), or a part of adjacent light converging grooves may be in contact with each other and the other part of adjacent light converging grooves are arranged at intervals. How to arrange the light converging grooves is not limited in the technical scheme of the present disclosure.
(118) It is found through testing that when the light guide layer **340** in the front light source module is provided with the V-shaped light converging micro-structures **342** shown in FIG. **18**, the luminance of the display apparatus when presenting a gray scale L**255** is about 112.5 nit, and the luminance of the display apparatus when presenting a gray scale L0 is about 6.8 nit, that is, the contrast of the display apparatus is about 16.5. When the light guide layer **340** in the front light source module is not provided with the light converging micro-structures **342** (on a plane parallel to the first direction X and on the surface of the light guide layer **340** away from the first light adjusting layer **320**), the luminance of the display apparatus when presenting the gray scale L**255** is about 89 nit, and the luminance of the display apparatus when presenting the gray scale L0 is about 5.8 nit, that is, the contrast of the display apparatus is about 15.3. According to the above data, it is found that the light converging micro-structures **342** are disposed in the light guide layer **340**, so that the display brightness and the contrast of the display apparatus can be improved.
(119) In some embodiments, a length L3 of each light converging groove in the first direction X satisfies: L3≤80 μm.
(120) In the embodiment of the present disclosure, a size of each light converging groove is relatively small, and therefore it is difficult to form the light converging groove by a conventional patterning process. Therefore, in the embodiment of the present disclosure, a material of the light guide layer **340** is a nano-imprint material, and the light converging grooves with a small size may be formed by the nano-imprint process.
(121) FIG. **20** is a schematic diagram of another structure of a front light source module according to an embodiment of the present disclosure; FIG. **21** is a schematic cross-sectional view of another front light source module according to an embodiment of the present disclosure. As shown in FIG. **20** and FIG. **21**, in some embodiments, at least one second light adjusting layer **350** is provided on a side of the light guide layer **340** away from the first light adjusting layer **320**, the at least one second light adjusting layer **350** and the light guide layer **340** are stacked in the first direction X, and light adjusting micro-structures **360** are disposed on the second light adjusting layer **350** and configured to adjust an exit angle of light emitted from the surface of the light guide layer **340** away from the first light adjusting layer **320** and passing through the light adjusting micro-structures **360**.
(122) FIG. **22** is a schematic diagram illustrating modulation of light by three different light adjusting micro-structures according to an embodiment of the present disclosure. As shown in FIG. **22**, the light emitted from the light guide layer **340** away from the first light adjusting layer **320** propagates in a direction away from the plane where the light incident side **341** is located. According to the different light adjusting functions, the light adjusting micro-structures in the present disclosure may be divided into a first light adjusting micro-structure **3601**, a second light adjusting micro-structure **3602**, and a third light adjusting micro-structure **3603**.
(123) The first light adjusting micro-structure **3601** is configured so that the light emitted from the surface of the light guide layer **340** away from the first light adjusting layer **320** and passing through the first light adjusting micro-structure **3601** still propagates in the direction away from the plane where the light incident side **341** is located, but an angle between the light and the third direction Z increases.
(124) The second light adjusting micro-structure **3602** is configured so that the light emitted from the surface of the light guide layer **340** away from the first light adjusting layer **320** and passing through the second light adjusting micro-structure **3602** still propagates in the direction away from the plane where the light incident side **341** is located, but the angle between the light and the third

direction Z decreases.
(125) The third light adjusting micro-structure **3603** is configured so that the light emitted from the surface of the light guide layer **340** away from the first light adjusting layer **320** and passing through the third light adjusting micro-structure **3603** propagates in a direction close to the plane where the light incident side **341** is located.
(126) In some embodiments, a cross section of each light adjusting micro-structure taken along a plane perpendicular to the second direction Y is triangular. At this time, the entire light adjusting micro-structure has a triangular prism shape, and an extending direction of the light adjusting micro-structure is parallel to the second direction Y. The triangular prism shape includes a third inclined surface **362** and a fourth inclined surface **361** intersecting with the first direction X, and the third inclined surface **362** is closer to the plane where the light incident side **341** is located than the fourth inclined surface **361**. As can be seen from the optical path in FIG. **22**, the fourth inclined surface **361** is configured to adjust the exit angle of the light.
(127) An angle e1 between the fourth inclined surface **361** of the first light adjusting micro-structure **3601** and the first direction X is smaller than an angle e2 between the fourth inclined surface **361** of the second light adjusting micro-structure **3602** and the first direction X, and the angle e2 between the fourth inclined surface **361** of the second light adjusting micro-structure **3602** and the first direction X is smaller than an angle e3 between the fourth inclined surface **361** of the third light adjusting micro-structure **3603** and the first direction X. An angle between the third inclined surface **362** in each of the first light adjusting micro-structure **3601**, the second light adjusting micro-structure **3602**, and the third light adjusting micro-structure **3603** and the first direction X is not limited in the present disclosure. As an alternative, an angle f1 between the third inclined surface **362** in the first light adjusting micro-structure **3601** and the first direction X is greater than an angle f2 between the third inclined surface **362** in the second light adjusting micro-structure **3602** and the first direction X, and the angle f2 between the third inclined surface **362** in the second light adjusting micro-structure **3602** and the first direction X is greater than an angle f3 between the third inclined surface **362** in the third light adjusting micro-structure **3603** and the first direction X.
(128) In the embodiment of the present disclosure, the number of the at least one second light adjusting layer **350** may be 1, 2, 3 or more, and at least one of the first light adjusting micro-structure **3601**, the second light adjusting micro-structure **3602** and the third light adjusting micro-structure **3603** may be selectively disposed on each second light adjusting layer **350**. The number of the second light adjusting layers **350** and the types of the light adjusting micro-structures disposed on each second light adjusting layer **350** are not limited in the technical solution of the present disclosure.
(129) In some embodiments, a refractive index of the second light adjusting layer **350** closest to the light guide layer **340** is greater than or equal to the refractive index of the light guide layer **340**.
(130) In some embodiments, the number of the second light adjusting layers **350** is greater than or equal to 2. A refractive index of one of any two adjacent second light adjusting layers **350** closer to the light guide layer **340** is less than or equal to a refractive index of the other second light adjusting layer **350**.
(131) In some embodiments, a first adhesive layer (not shown) is disposed between the second light adjusting layer **350** closest to the light guide layer **340** and the light guide layer **340**, and has a refractive index greater than or equal to that of the light guide layer **340** and less than or equal to that of the second light adjusting layer **350** in contact with the first adhesive layer. A second adhesive layer (not shown) is disposed between any two adjacent second light adjusting layers **350**, and a refractive index of each second adhesive layer is greater than or equal to a refractive index of the second light adjusting layer **350** in contact with a surface of the second adhesive layer close to the light guide layer **340** and is less than or equal to a refractive index of the second light adjusting layer **350** in contact with a surface of the second adhesive layer away from the light guide layer

340. In this way, the light emitted from the light guide layer **340** toward the reflective display panel **100** is prevented from being totally reflected during the propagation process, so as to ensure the quantity of light reaching the reflective display panel **100**.
(132) As an example, in the embodiment of the present disclosure, a thickness of the light guide layer **340** is in a range from 0.2 mm to 0.4 mm, a thickness of the first light adjusting layer **320** is in a range from 0.085 mm to 0.145 mm, a thickness of the first attaching adhesive layer **330** is in a range from 0.05 mm to 0.1 mm, and a thickness of the second light adjusting layer **350** is in a range from 0 mm to 0.2 mm.
(133) In the actual production process, the light guide layer **340** and the first light adjusting layer **320** may be formed respectively. The micro-groove structures **310** in the first light adjusting layer **320** may be formed by a nano-imprint process (generally including molding, imprinting, de-molding and the like). If the light guide layer **340** is provided with the light converging micro-structures, the light converging micro-structures may be formed in the light guide layer **340** through a nano-imprint process, and the light guide layer **340** is then attached and fixed to the first light adjusting layer **320** by the first attaching adhesive layer **330**.
(134) Then, the second light adjusting layer **350** is selectively formed according to the requirement, and the second light adjusting layer **350** is fixed to the light guide layer **340**.
(135) Based on the same concept, the embodiment of the present disclosure further provides a display apparatus. Referring to FIG. **1**, the display apparatus includes: a reflective display panel **100** and a front light source module **300**, the front light source module **300** is located on a light outgoing surface of the reflective display panel **100**, the front light source module **300** is similar to the front light source module **300** provided in the previous embodiments. For the detailed description of the front light source module **300**, reference may be made to the contents in the previous embodiments, which is not repeated herein.
(136) In some embodiments, the reflective display panel **100** includes a plurality of sub-pixel regions arranged in an array along the first direction X and the second direction Y, each sub-pixel region has a length L.sub.0 in the first direction X; a length of each micro-groove structure **310** in the first direction X is less than or equal to ⅔×L.sub.0, and a length of each micro-groove structure **310** in the second direction Y is less than or equal to ⅔×L.sub.0. With this arrangement, each micro-groove structure **310** has a small size, and the micro-groove structures **310** are invisible when the display apparatus is watched (on the front) by human eyes, so that the display effect of the display apparatus is improved.
(137) In some embodiments, the plurality of light converging micro-structures are provided in the surface of the light guide layer **340** away from the first light adjusting layer **320** (see the content in the foregoing embodiments), and are configured to converge the light passing through the light converging micro-structures from the light guide layer **340**. A length of each light converging micro-structure in the first direction X is less than or equal to ⅔×L.sub.0. With this arrangement, each light converging micro-structure has a small size, and the light converging micro-structures are invisible when the display apparatus is watched (on the front) by human eyes, so that the display effect of the display apparatus is improved.
(138) In some embodiments, at least one second light adjusting layer **350** is provided on a side of the light guide layer **340** away from the first light adjusting layer **320**, at least one second light adjusting layer **350** and the light guide layer **340** are stacked in the first direction X, and the light adjusting micro-structures **360** (see the content in the foregoing embodiments) are disposed on the second light adjusting layer **350** and configured to adjust an exit angle of light emitted from the surface of the light guide layer **340** away from the first light adjusting layer **320** and passing through the light adjusting micro-structures **360**. A length of each light adjusting micro-structure in the first direction X is less than or equal to ⅔×L.sub.0. With this arrangement, each light adjusting micro-structure has a small size, and the light adjusting micro-structures are invisible when the display apparatus is watched (on the front) by human eyes, so that the display effect of the display

apparatus is improved.
(139) As an example, when a size of the reflective display panel **100** is 1.5 inches, the length L.sub.0 of each sub-pixel region in the first direction X is generally 36 μm. When the size of the reflective display panel **100** is 8 inches, the length L.sub.0 of each sub-pixel region in the first direction X is generally 60 μm. When the size of the reflective display panel **100** is 30 inches, the length L.sub.0 of each sub-pixel region in the first direction X is generally 80 μm.
(140) FIG. **23** is a schematic cross-sectional view of a portion of a display apparatus according to an embodiment of the present disclosure. As shown in FIG. **23**, in the embodiment of the present disclosure, by means of the double-layer design of the light guide layer **340** and the first light adjusting layer **320** in the present disclosure, a thickness of the light guide layer **340** (in a range from 0.2 mm to 0.4 mm) may be smaller than a thickness of a single-layer light guide plate (generally greater than 0.4 mm, and about 0.8 mm) in the related art, and due to the reduction of the thickness of the light guide layer **340**, a size of the light emitting element disposed on the light guide layer **340** (generally, a length of the light emitting element in the third direction Z is slightly smaller than a length of the light incident side **341** of the light guide layer **340** in the third direction Z) may be correspondingly reduced, so that a distance between two adjacent light emitting elements on a driving backplane may be reduced, and a light mixing distance (a distance from the light emitting element to an effective display region AA of the display apparatus) required by the light emitting elements arranged at intervals may be correspondingly reduced.
(141) In the related art, each light emitting element disposed on the light guide plate with a great thickness have a length of about 1.7 mm and a width of about 1.7 mm, and a required minimum light mixing distance is about 3.6 mm. In the present disclosure, each light emitting element disposed on the light guide layer **340** with a smaller thickness have a length less than 0.56 mm and a width less than 0.56 mm, and a required minimum light mixing distance is about 1.5 mm.
(142) FIG. **24** is another schematic cross-sectional view of a portion of a display apparatus according to an embodiment of the present disclosure. As shown in FIG. **24**, the display apparatus may include not only the reflective display panel **100** and the front light source module **300**, but also a touch substrate **500**, wherein the touch substrate **500** is located on a side of the front light source module **300** away from the reflective display panel **100**, and is fixed to the front light source module **300** through a foam tape **700**, so that the display apparatus has a touch function.
(143) In addition, in some embodiments, a reflection tape **600** may be used to fix the side light source **400** and the light guide layer **340**, so that the side light source **400** and the light guide layer **340** may be fixed, and it can be ensured that the light cannot be emitted from an upper surface and a lower surface of the light guide layer **340** in the light mixing process, thereby improving the light utilization rate.
(144) In some embodiments, the front light source module **300** is attached to the reflective display panel **100** through a second attaching adhesive layer **200**, and a refractive index of a portion of the front light source module **300** in contact with the second attaching adhesive layer **200** is greater than a refractive index of the second attaching adhesive layer **200**. For example, when the front light source module **300** does not include the second light adjusting layer **350**, the light guide layer **340** is fixed to the reflective display panel **100** through the second attaching adhesive layer **200**. When the front light source module **300** includes the second light adjusting layer **350**, the second light adjusting layer **350** is fixed to the reflective display panel **100** through the second attaching adhesive layer **200**.
(145) It should be understood that the above embodiments are merely exemplary embodiments adopted to explain the principles of the present disclosure, and the present disclosure is not limited thereto. It will be apparent to one of ordinary skill in the art that various changes and modifications may be made therein without departing from the spirit and scope of the present disclosure, and such changes and modifications also fall within the scope of the present disclosure.

Claims

1. A front light source module, comprising: a side light source; a light guide layer comprising a light incident side opposite to the side light source in a first direction; and a first light adjusting layer, wherein the first light adjusting layer and the light guide layer are stacked in a third direction, a side of the first light adjusting layer away from the light guide layer is provided with a plurality of micro-groove structures, each micro-groove structure comprises: a first inclined surface and a second inclined surface opposite to each other in the first direction, the first inclined surface is configured to face the light incident side and is closer to the light incident side than the second inclined surface, an angle α between the first inclined surface and a plane where a surface of the first light adjusting layer away from the light guide layer is located is in a range from 26° to 42°, and a depth H of each micro-groove structure is in a range from 4 μm to 15 μm; wherein a refractive index of the first light adjusting layer is greater than or equal to that of the light guide layer; wherein a distribution density of the plurality of micro-groove structures increases gradually in a direction away from the light incident side along the first direction; wherein the surface of the first light adjusting layer away from the light guide layer is divided into a plurality of groove structure arrangement regions arranged along the first direction; a distance between every two adjacent groove structure arrangement regions gradually decreases in the direction away from the light incident side along the first direction; each groove structure arrangement region is divided into a plurality of rectangular periodic regions arranged in a second direction; a length of each rectangular periodic region in the first direction is R, and a length of each rectangular periodic region in the second direction is Q, and the second direction is perpendicular to both the first direction and the third direction; M micro-groove structures are uniformly arranged in each rectangular periodic region; and wherein the M micro-groove structures are arranged in each rectangular periodic region to satisfy: a distance between centers of any two micro-groove structures in the first direction is greater than or equal to R/M, and a distance between centers of any two micro-groove structures in the second direction is greater than or equal to Q/M.
2. The front light source module of claim 1, wherein a length of an opening of each micro-groove structure in the surface of the first light adjusting layer away from the light guide layer in the first direction is L.sub.1; and a ratio of L1 to H is L1/H, which satisfies: L.sub.1/H≤4.
3. The front light source module of claim 2, wherein L.sub.1 satisfies L.sub.1≤80 μm.
4. The front light source module of claim 1, wherein the first inclined surface is rectangular, a pair of opposite sides of the rectangular first inclined surface extend along a second direction, and the second direction is perpendicular to both the first direction and the third direction; and an extending direction of the other pair of opposite sides of the rectangular first inclined surface is perpendicular to the third direction, and intersects with both the first direction and the second direction; and a length L.sub.2 of a side of the rectangular first inclined surface extending in the second direction satisfies: L.sub.2≤80 μm.
5. The front light source module of claim 1, wherein the first light adjusting layer is attached to the light guide layer by a first attaching adhesive layer, and a refractive index of the first attaching adhesive layer is greater than or equal to the refractive index of the light guide layer and is less than or equal to a refractive index of the first light adjusting layer.
6. The front light source module of claim 1, wherein a shape of a cross section of each micro-groove structure taken along a plane parallel to the first direction and parallel to the third direction comprises: a triangle or a quadrilateral; and wherein the second inclined surface is a flat surface or a curved surface.
7. The front light source module of claim 1, wherein the side light source comprises: a light source; and a converging structure between the light source and the light incident side and configured to converge the light emitted from the light source, wherein an angle θ1 between light emitted from

the converging structure and a first reference plane satisfies: θ1≤52.4°, and the first reference plane is a plane perpendicular to the third direction.
8. The front light source module of claim 7, wherein the converging structure comprises: a wedge-shaped light guiding structure; the wedge-shaped light guiding structure comprises: a first light incident surface, a first light outgoing surface, a first light adjusting surface and a second light adjusting surface; the first light incident surface and the first light outgoing surface are opposite to each other in the first direction, the first light incident surface and the first light outgoing surface are perpendicular to the first direction, a length of the first light incident surface in the third direction is T1, a length of the first light outgoing surface in the third direction is T2, T2>T1, and an orthographic projection of the first light outgoing surface on a plane where the first light incident surface is located covers the first light incident surface; the first light adjusting surface and the second light adjusting surface are opposite to each other in the third direction, and a distance between the first light adjusting surface and the second light adjusting surface in the third direction is gradually increased in a direction from the first light incident surface to the first light outgoing surface along the first direction; and the light source is opposite to the first light incident surface, and the light incident side is opposite to the first light outgoing surface.
9. The front light source module of claim 8, wherein the light source comprises: a driving board and a light emitting element fixed onto the driving board, a length T of the light emitting element in the third direction is smaller than the length T1 of the first light incident surface in the third direction; and an orthographic projection of the light emitting element on the plane where the first light incident surface is located in a region defined by the first light incident surface; the length T of the light emitting element in the third direction satisfies: T≤0.3 mm.
10. The front light source module of claim 7, wherein the converging structure comprises a condenser lens; a shape of a cross section, taken along a plane perpendicular to the second direction, of a surface of the condenser lens away from the light source is a circular arc; or a shape of a cross section, taken along a plane perpendicular to the second direction, of a surface of the condenser lens away from the light source is a curve formed by connecting circular arcs and line segments sequentially and alternately; and the light source comprises: a driving board and a light emitting element fixed onto the driving board, and the condenser lens is arranged on the driving board and covers the light emitting element.
11. The front light source module of claim 1, further comprising a plurality of light converging micro-structures on a surface of the light guide layer away from the first light adjusting layer, and configured to converge light passing through the plurality of light converging micro-structures from the light guide layer.
12. The front light source module of claim 11, wherein each light converging micro-structure is a light converging groove on the surface of the light guide layer away from the first light adjusting layer, and the light converging groove extends along the second direction; and a cross section of a surface of the light converging groove taken along a plane perpendicular to the second direction Y is V-shaped or arc-shaped; and a length L.sub.3 of the light converging groove in the first direction satisfies: L.sub.3≤80 μm.
13. The front light source module of claim 8, wherein the wedge-shaped light guiding structure and the light guide layer are made of a same material and have a one-piece structure; and the light incident side and the first light outgoing surface are a same surface.
14. A front light source module, comprising: a side light source; a light guide layer comprising a light incident side opposite to the side light source in a first direction; and a first light adjusting layer, wherein the first light adjusting layer and the light guide layer are stacked in a third direction, a side of the first light adjusting layer away from the light guide layer is provided with a plurality of micro-groove structures, each micro-groove structure comprises: a first inclined surface and a second inclined surface opposite to each other in the first direction, the first inclined surface is configured to face the light incident side and is closer to the light incident side than the second

inclined surface, an angle a between the first inclined surface and a plane where a surface of the first light adjusting layer away from the light guide layer is located is in a range from 26° to 42°, and a depth H of each micro-groove structure is in a range from 4 μm to 15 μm; wherein a refractive index of the first light adjusting layer is greater than or equal to that of the light guide layer; the front light source module further comprises at least one second light adjusting layer on a side of the light guide layer away from the first light adjusting layer, wherein the at least one second light adjusting layer and the light guide layer are stacked in the third direction, and the at least one second light adjusting layer comprises light adjusting micro-structures thereon and configured to adjust an exit angle of light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the light adjusting micro-structures; wherein a refractive index of a second light adjusting layer of the at least one second light adjusting layer closest to the light guide layer is greater than or equal to a refractive index of the light guide layer; and wherein the at least one second light adjusting layer comprises two or more second light adjusting layers; and a refractive index of one of any two adjacent second light adjusting layers closer to the light guide layer is less than or equal to a refractive index of the other second light adjusting layer; and wherein the light emitted from the surface of the light guide layer away from the first light adjusting layer propagates in a direction away from a plane where the light incident side is located; the light adjusting micro-structures arranged on the at least one second light adjusting layer comprise: a first light adjusting micro-structure configured to adjust the light so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the first light adjusting micro-structure still propagates in the direction away from the plane where the light incident side is located, but an angle between the light and the third direction increases; and/or the light adjusting micro-structures arranged on the at least one second light adjusting layer comprise: a second light adjusting micro-structure configured to adjust the light so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the second light adjusting micro-structure still propagates in the direction away from the plane where the light incident side is located, but the angle between the light and the third direction decreases; and/or the light adjusting micro-structures arranged on the at least one second light adjusting layer comprise: a third light adjusting micro-structure configured to adjust the light so that the light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the third light adjusting micro-structure propagates in a direction close to the plane where the light incident side is located.

15. A display apparatus, comprising: a reflective display panel and a front light source module, wherein the front light source module is on a light outgoing surface of the reflective display panel; the front light source module comprises: a side light source; a light guide layer comprising a light incident side opposite to the side light source in a first direction; and a first light adjusting layer, wherein the first light adjusting layer and the light guide layer are stacked in a third direction, a side of the first light adjusting layer away from the light guide layer is provided with a plurality of micro-groove structures, each micro-groove structure comprises: a first inclined surface and a second inclined surface opposite to each other in the first direction, the first inclined surface is configured to face the light incident side and is closer to the light incident side than the second inclined surface, an angle a between the first inclined surface and a plane where a surface of the first light adjusting layer away from the light guide layer is located is in a range from 26° to 42°, and a depth H of each micro-groove structure is in a range from 4 μm to 15 μm; wherein a refractive index of the first light adjusting layer is greater than or equal to that of the light guide layer; and wherein the reflective display panel comprises a plurality of sub-pixel regions arranged in an array along a first direction and a second direction, and each sub-pixel region has a length L.sub.0 in the first direction; and a length of each micro-groove structure in the first direction is less than or equal to ⅔×L.sub.0, and a length of each micro-groove structure in the second direction is less than or equal to ⅔×L.sub.0; the display apparatus further comprises a plurality of light

converging micro-structures on a surface of the light guide layer away from the first light adjusting layer, and configured to converge light passing through the plurality of light converging micro-structures from the light guide layer; wherein a length of each light converging micro-structure in the first direction is less than or equal to ⅔×L.sub.0; and the display apparatus further comprises at least one second light adjusting layer on a side of the light guide layer away from the first light adjusting layer, wherein the at least one second light adjusting layer and the light guide layer are stacked in the first direction, and the at least one second light adjusting layer comprises light adjusting micro-structures thereon and configured to adjust an exit angle of light emitted from the surface of the light guide layer away from the first light adjusting layer and passing through the light adjusting micro-structures; wherein a length of each light adjusting micro-structure in the first direction is less than or equal to ⅔×L.sub.0.

16. The display apparatus of claim 15, wherein the front light source module and the reflective display panel are attached to each other by a second attaching adhesive layer; and a refractive index of a portion of the front light source module in contact with the second attaching adhesive layer is greater than a refractive index of the second attaching adhesive layer.